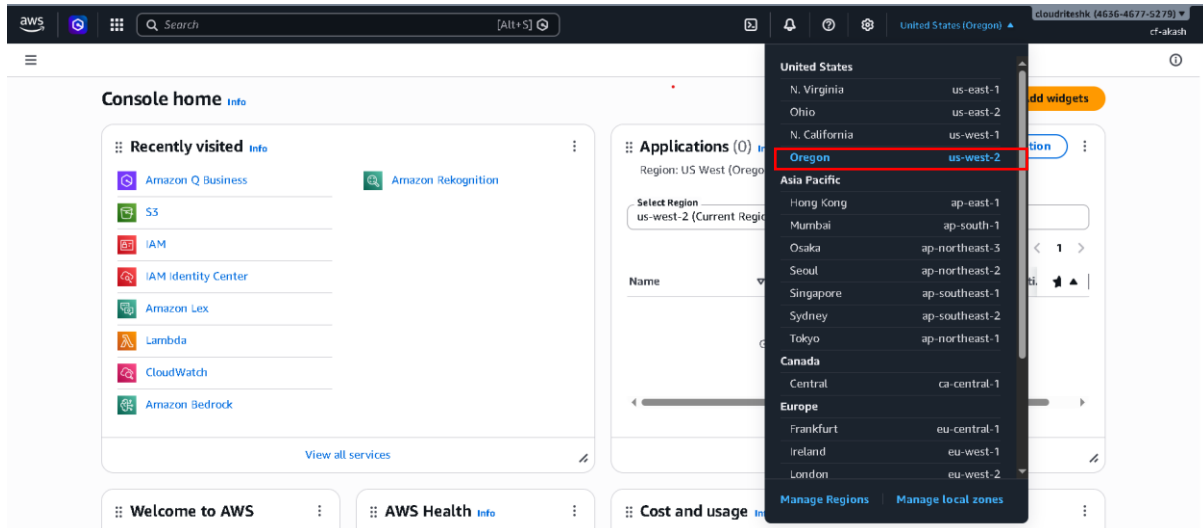


Amazon Bedrock Knowledge Base: -

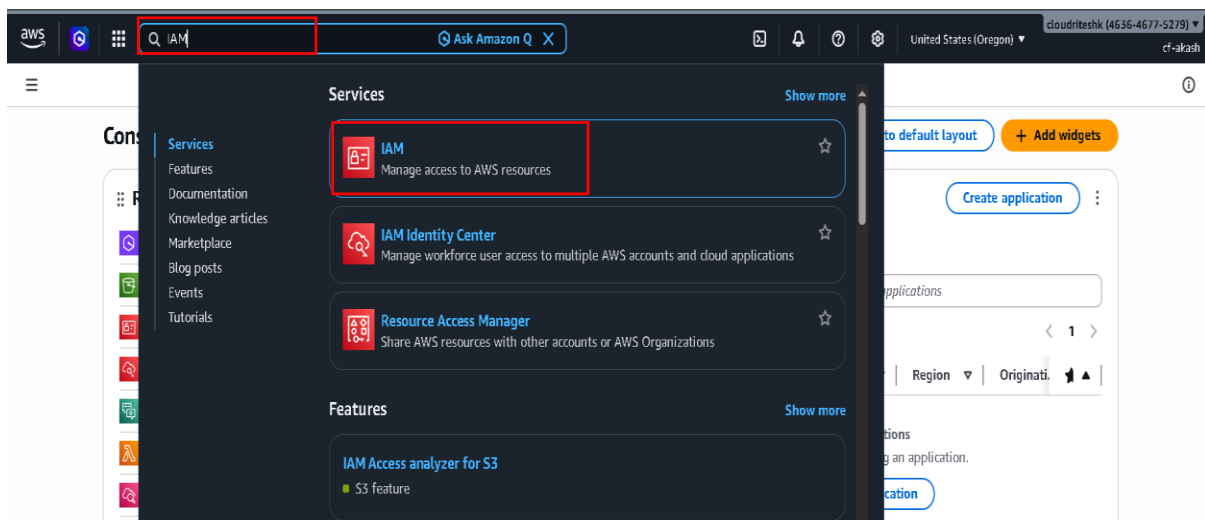
Summary: -

I built a smart AI system using Amazon Bedrock that reads your uploaded documents from Amazon S3 and answers user questions based on that data. It works like a personal assistant that searches your files and gives accurate, source-based responses.

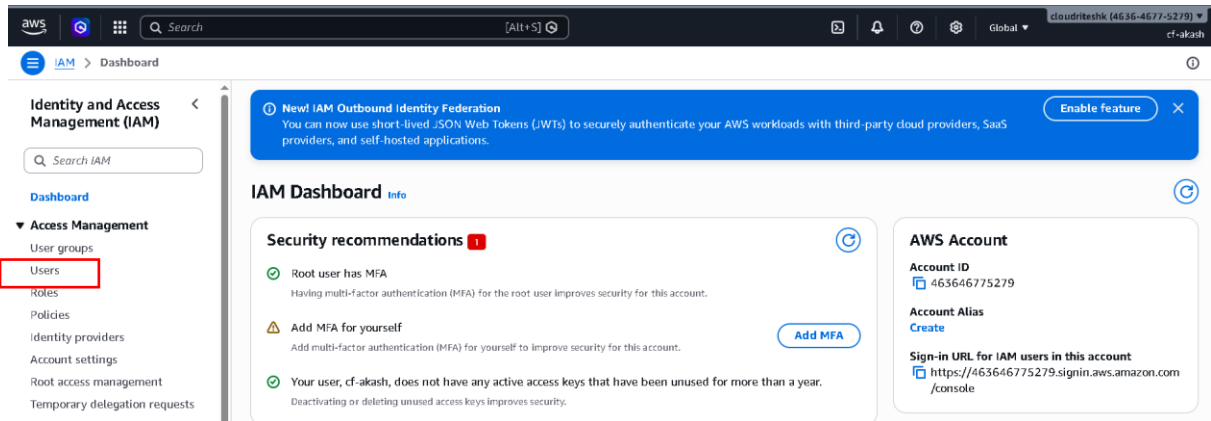
- Before you start please shift to region us-west-2.
- Because some of the services not work in other regions.



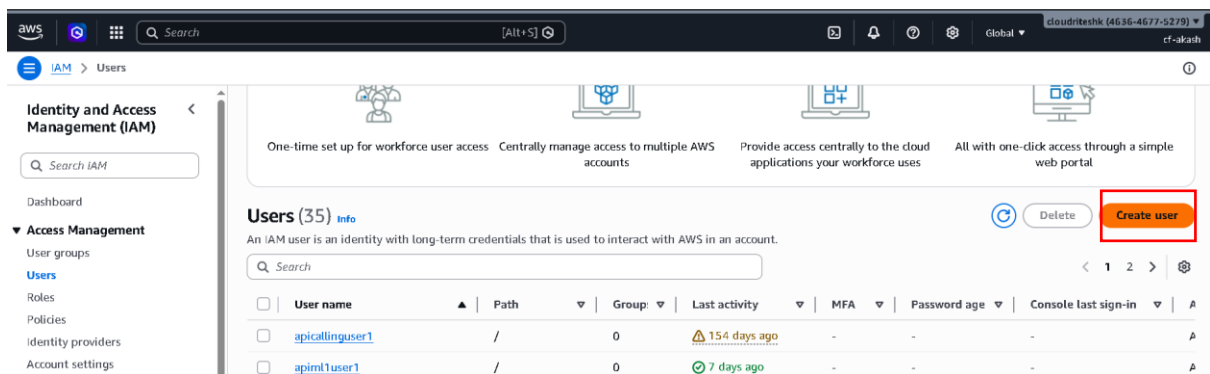
- Second thing you can't create Amazon Bedrock Knowledge Base if you logged in as root user



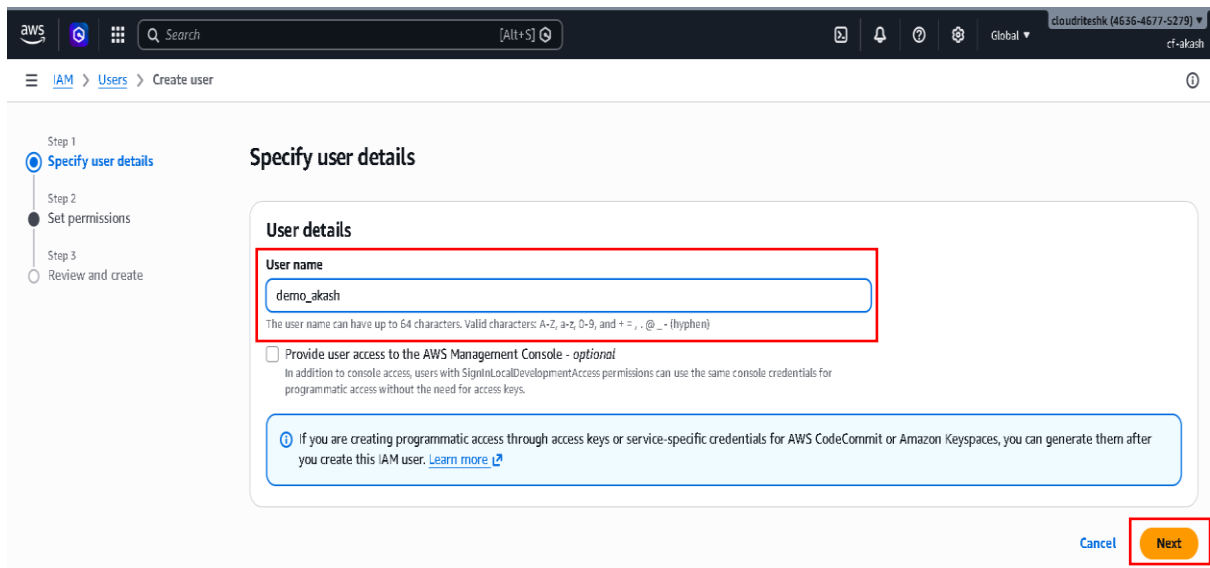
- So what you will have to do is you will have to create an IAM user.
- How you create IAM user: -
 - Go and search IAM in AWS console Search bar
- Now click on IAM



- Now click on user



- Now click on create user



- Enter User name and Enter Next.

Set permissions
Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.
- ☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/2344)
Choose one or more policies to attach to your new user. [Create policy](#)

Filter by Type: All types

Policy name	Type	Attached entities
aai_geforces3limitedpolicy	Customer managed	1
AccessAnalyzerServiceRolePolicy	AWS managed	0
AccountManagementFromVeracel	AWS managed	1
☒ AdministratorAccess	AWS managed - job function	40

- Select Permission options: -
 - Attach policies directly
- Add Permission policies: -
 - AdministratorAccess
- Now just click Next

Set permissions

☐	AiDevOpsAgentReadOnlyAccess	AWS managed	0
☐	AiDevOpsOperatorAppAccessPolicy	AWS managed	0
☐	AiOpsAssistantIncidentReportPolicy	AWS managed	0
☐	AiOpsAssistantPolicy	AWS managed	0
☐	AiOpsConsoleAdminPolicy	AWS managed	0
☐	AiOpsOperatorAccess	AWS managed	0
☐	AiOpsReadOnlyAccess	AWS managed	0
☐	AlexaForBusinessDeviceSetup	AWS managed	0
☐	AlexaForBusinessFullAccess	AWS managed	0
☐	AlexaForBusinessGatewayExecution	AWS managed	0
☐	AlexaForBusinessLifsizeDelegatedAc...	AWS managed	0

► Set permissions boundary - optional

[Cancel](#) [Previous](#) [Next](#)

Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name demo_akash	Console password type None	Require password reset No
-------------------------	-------------------------------	------------------------------

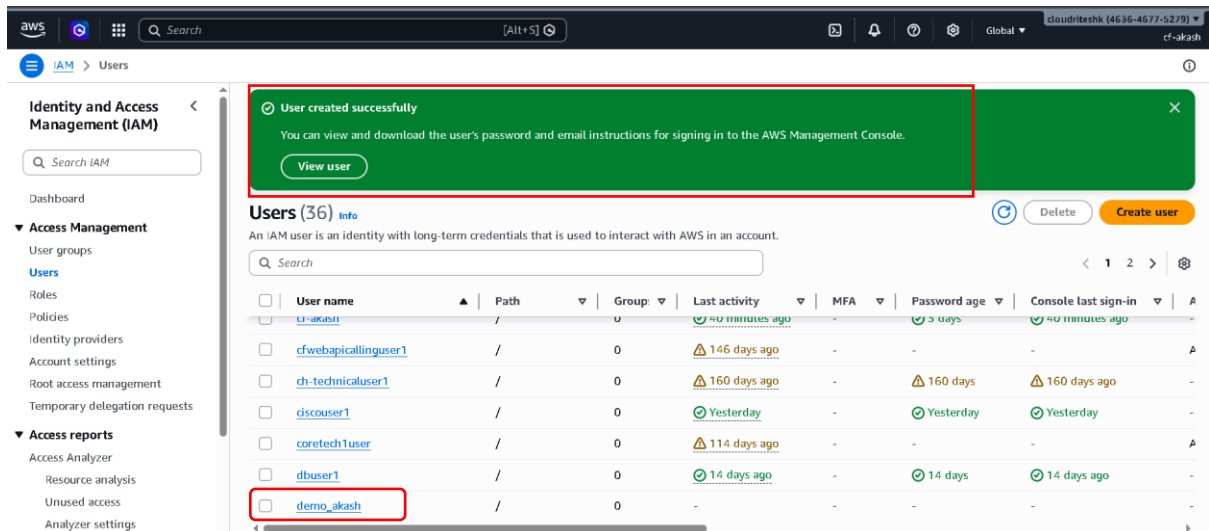
Permissions summary

Name	Type	Used as
AdministratorAccess	AWS managed - job function	Permissions policy

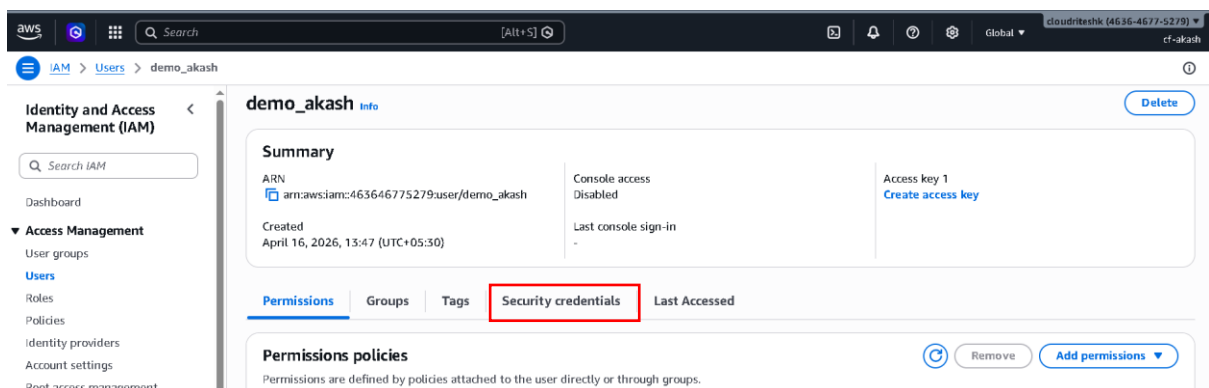
Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create user](#)

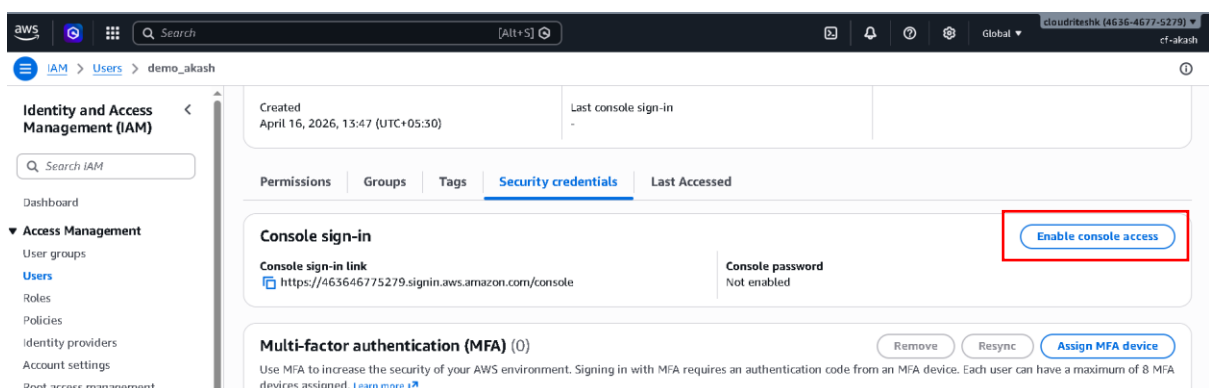
- Click create user.



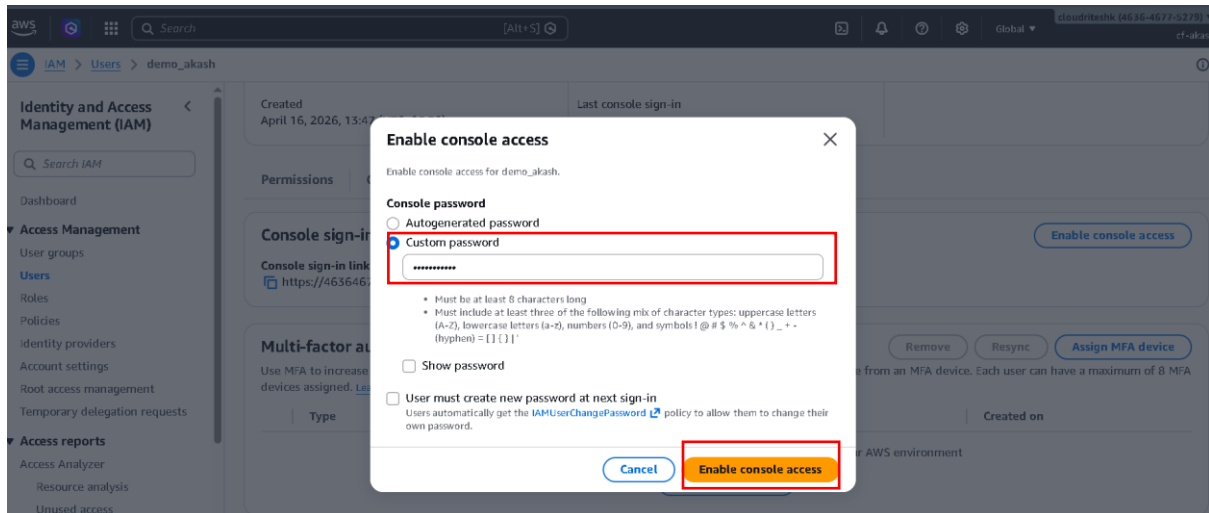
- Now our IAM user is created
- Click on the created user and go inside



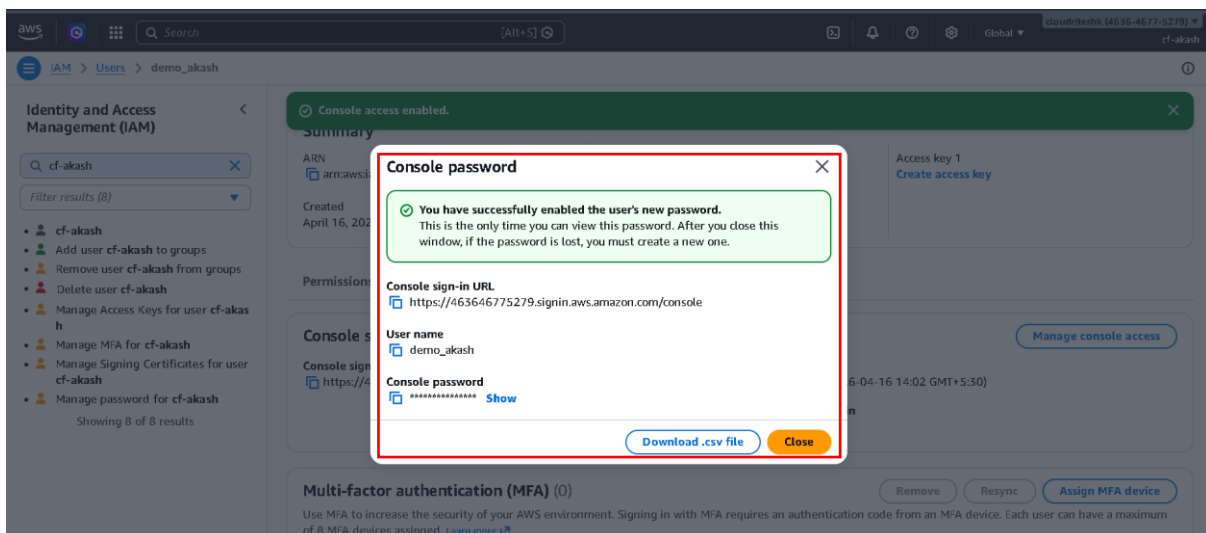
- Click on the security credentials.



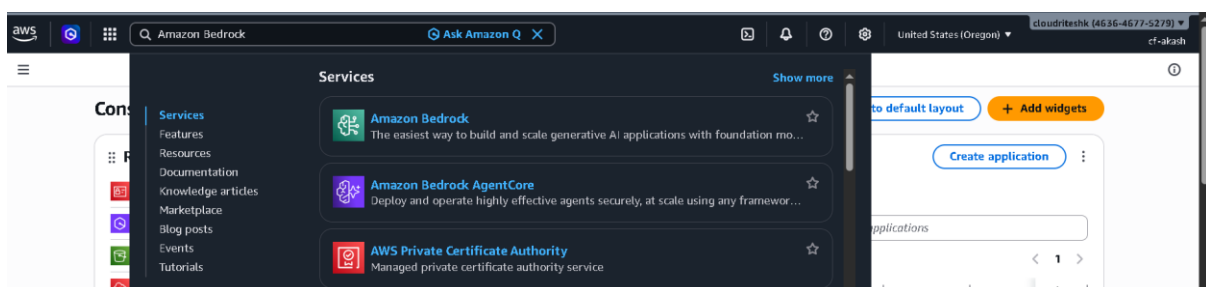
- Now just click Enable console access



- Now just Enter your password and click Enable console access

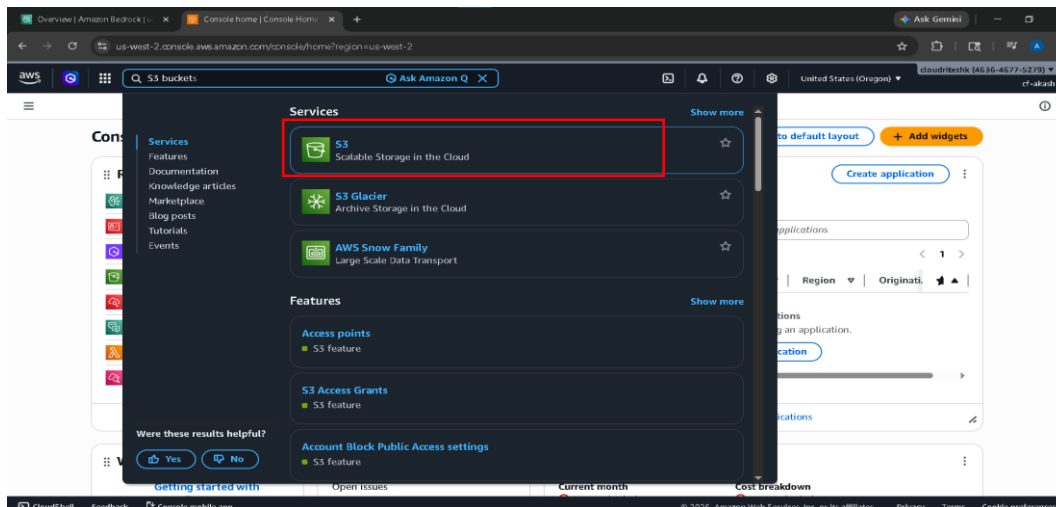


- Now just Sign out from root user and login from this IAM user that you have created.
- so now let's go and create our Amazon Bedrock Knowledge Base.

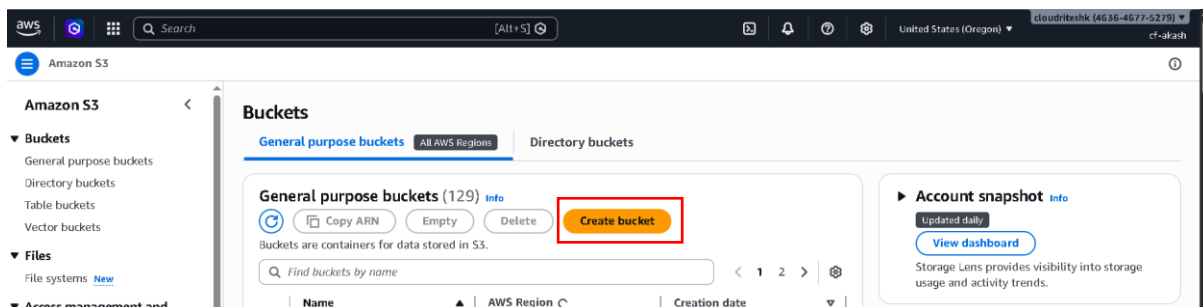


- Got to AWS console and search Amazon Bedrock.
- Click on Amazon Bedrock.

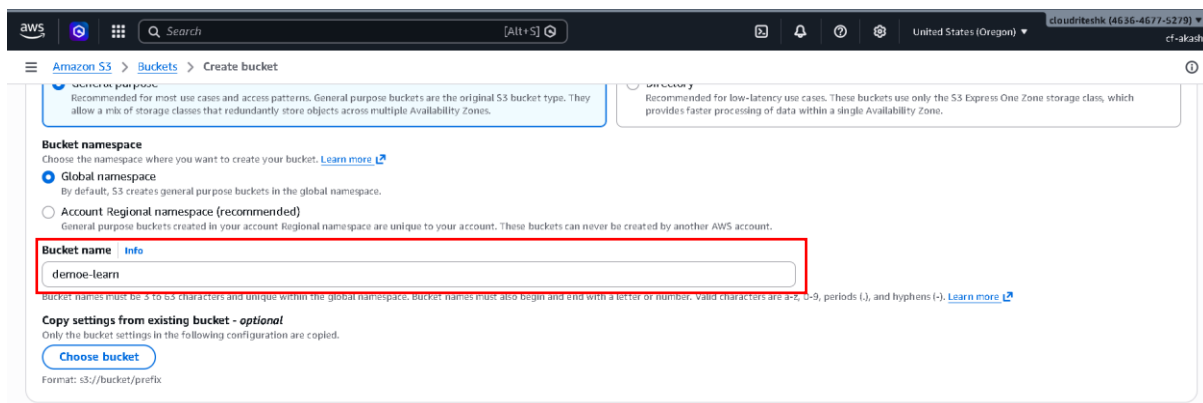
- The First thing for our E-learning application is
- Identifying Data source
- So we have to create an S3 bucket and upload all relative files which we want the user to be able to access as part of this E-learning platform.
- So let's create S3 Bucket first



- Open Aws in new tab and Search S3 bucket
- Click on S3



- Click Create bucket



- Enter Bucket name.

- Keep everything else default.

aws [Search] [Alt+S] United States (Oregon) cloudriteshk (4636-4677-5279) cf-akash

Amazon S3 > Buckets > Create bucket

You can add up to 50 tags.

Default encryption [Info](#)
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)
Secure your objects with two separate layers of encryption. For details on pricing, see [D5SE-KMS pricing](#) on the Storage tab of the [Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
- ☒ Enable

► **Advanced settings**

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel **Create bucket**

- Now just click create Button.

aws [Search] [Alt+S] United States (Oregon) cloudriteshk (4636-4677-5279) cf-akash

Amazon S3 > Buckets > demoe-learn

Successfully created bucket "demoe-learn"

demoe-learn [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects				

You don't have any objects in this bucket.

[Upload](#)

- Our S3 bucket is created.
- Now click on upload.

aws [Search] [Alt+S] United States (Oregon) cloudriteshk (4636-4677-5279) cf-akash

Amazon S3 > Buckets > demoe-learn > Upload

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (0) [Remove](#) [Add files](#) [Add folder](#)

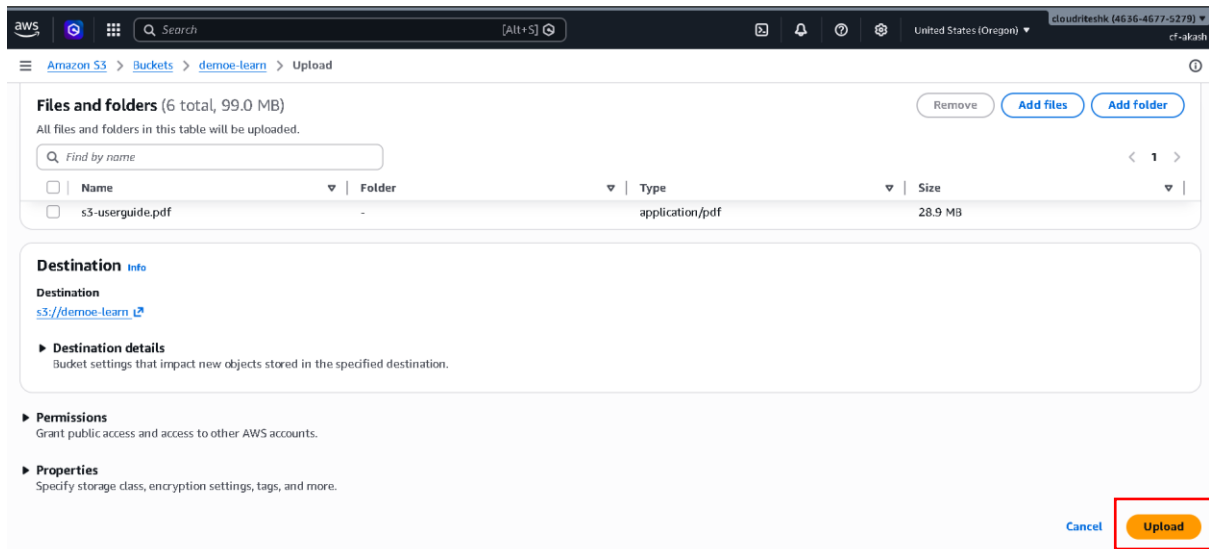
All files and folders in this table will be uploaded.

Find by name

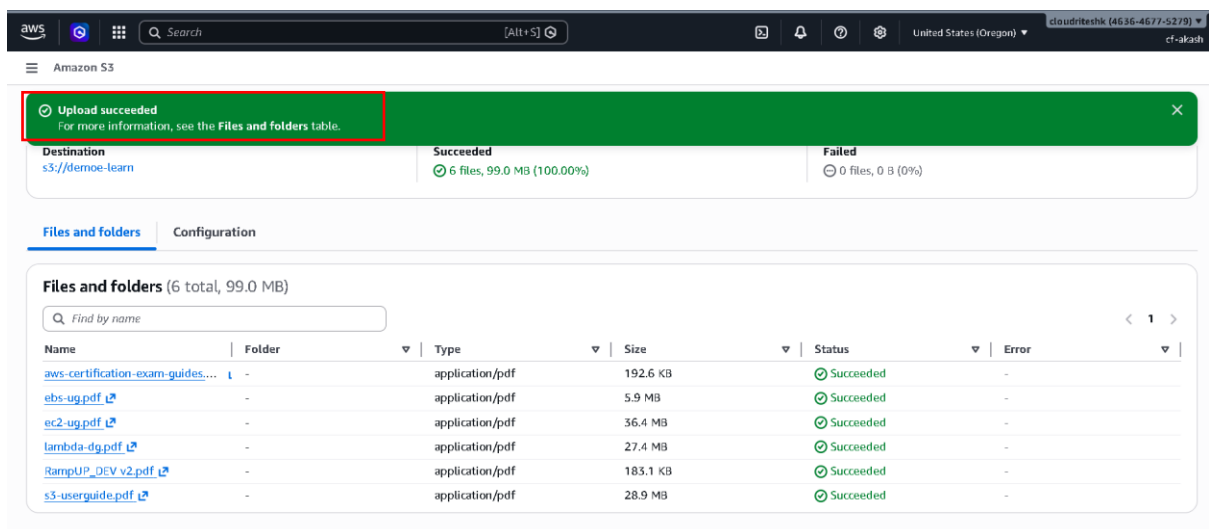
Name	Folder	Type	Size
No files or folders			

You have not chosen any files or folders to upload.

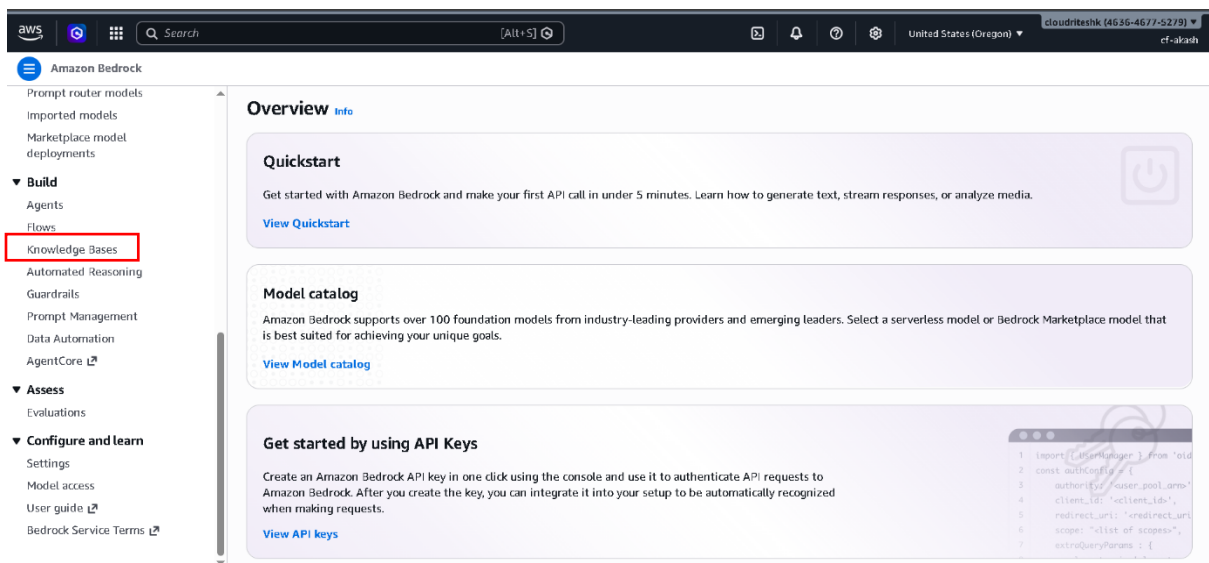
- Click Add files.
- And select files which we will use our internal data source.



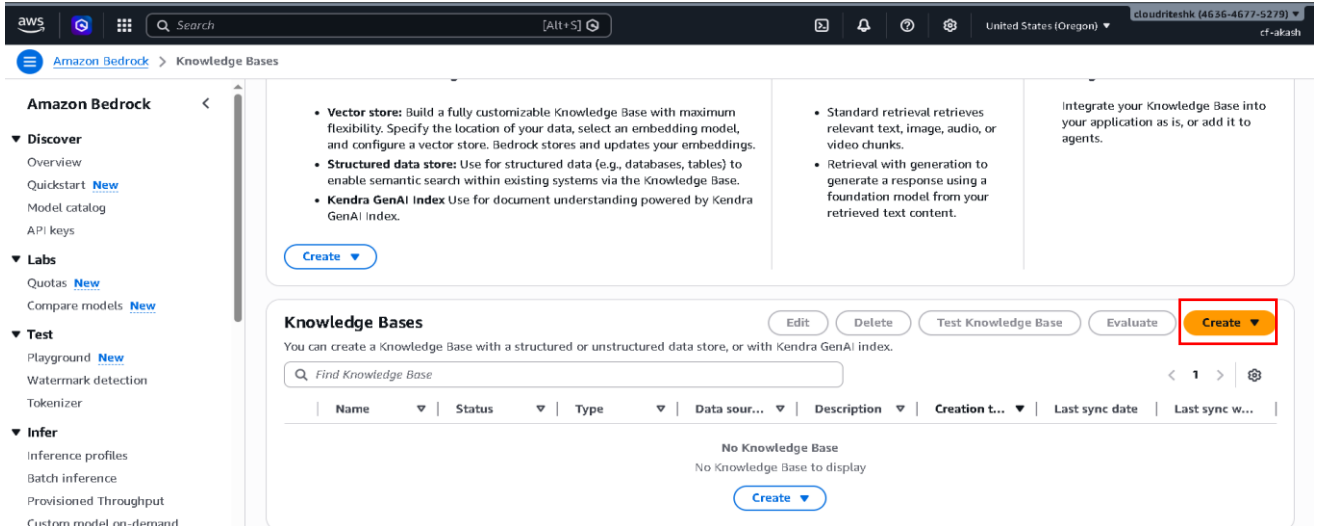
- Now just click upload



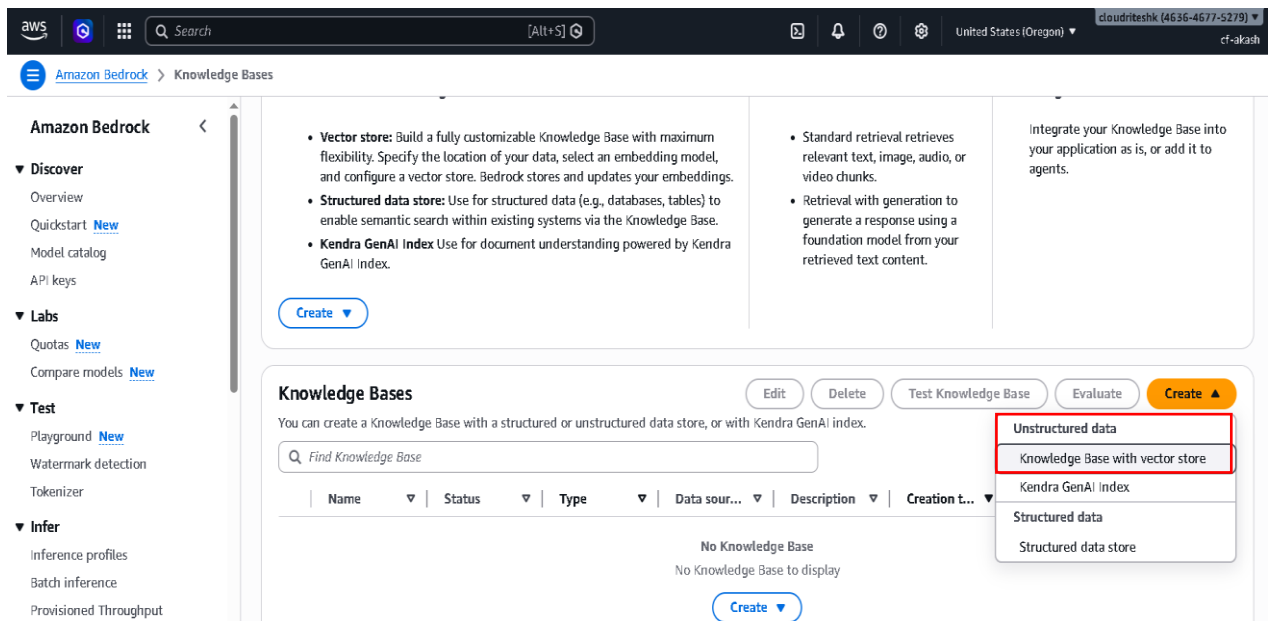
- Now our all Files is uploaded.
- Basically these files are AWS Documents like: -Amazon Ec2 guide, Lambda guide etc.
- Now Go to your Amazon Bedrock tab.



- Scroll down in navigation bar.
- Inside Build click knowledge Bases
- Before creating knowledge base I just want to warn you that's there going to be some cost involved.
- Because we are going to create a AWS open search server less vector store and that cost about \$0.50 per hour.



- Now just click on create.



- Choose option Knowledge Base with vector store.

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

Step 1: Provide Knowledge Base details

Knowledge Base details

Knowledge Base name
bedrock_knowledgebase
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 50 characters.

Knowledge Base description - optional
eLearning use case
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 200 characters.

IAM permissions
Certain permissions are necessary to access other services or perform actions in order to create this resource. For more information, see [service role](#) for Amazon Bedrock.

Runtime role

☒ Create and use a new service role
☐ Use an existing service role

Service role name

- Give Knowledge Base name
- Write Base description-optional
- IAM permission choose
 - Create and use a new service role.

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

Choose data source type
Select the data source type that you want to configure in the next step. You may add additional data sources once your Knowledge Base is created.

☒ **Amazon S3**
Object storage service that stores data as objects within general purpose buckets. Add up to 5 buckets as data sources.

☐ **Confluence - Preview**
Collaborative work-management tool designed for project planning, software development and product management.

☐ **Custom**
Create a data source directly in Amazon Bedrock. A custom data source allows the flexibility to automatically ingest documents into your vector database directly.

☐ **Salesforce - Preview**
Customer relationship management (CRM) tool for managing support, sales, and marketing data.

☐ **Sharepoint - Preview**
Collaborative web-based service for working on documents, web pages, web sites, lists, and more.

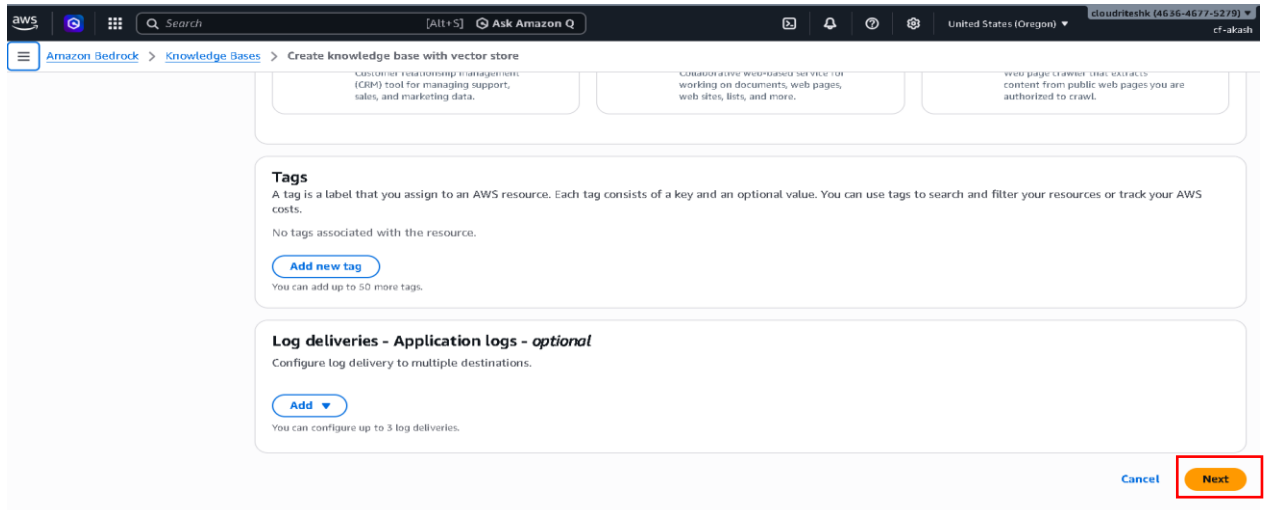
☐ **Web Crawler - Preview**
Web page crawler that extracts content from public web pages you are authorized to crawl.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

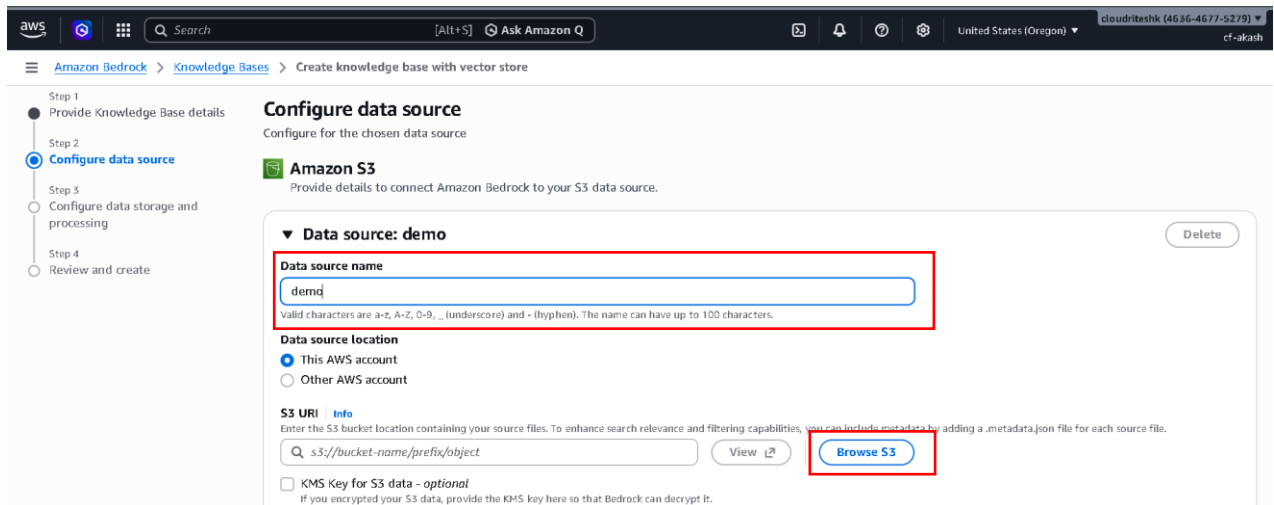
No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

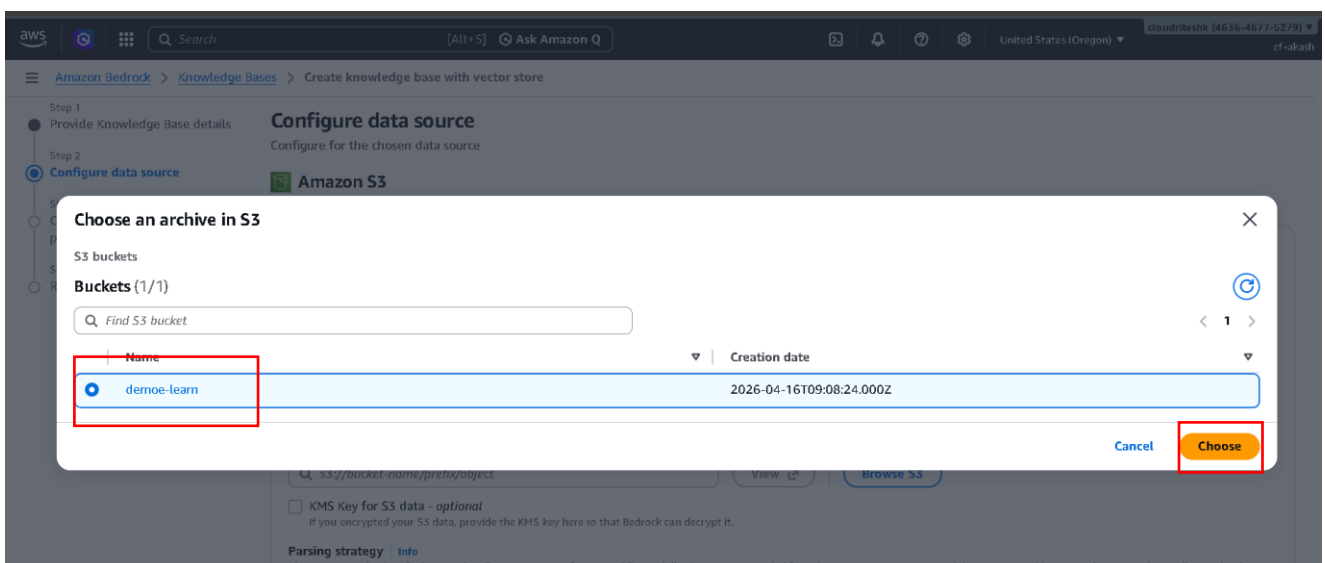
- Choose data source type:-
 - Amazon S3



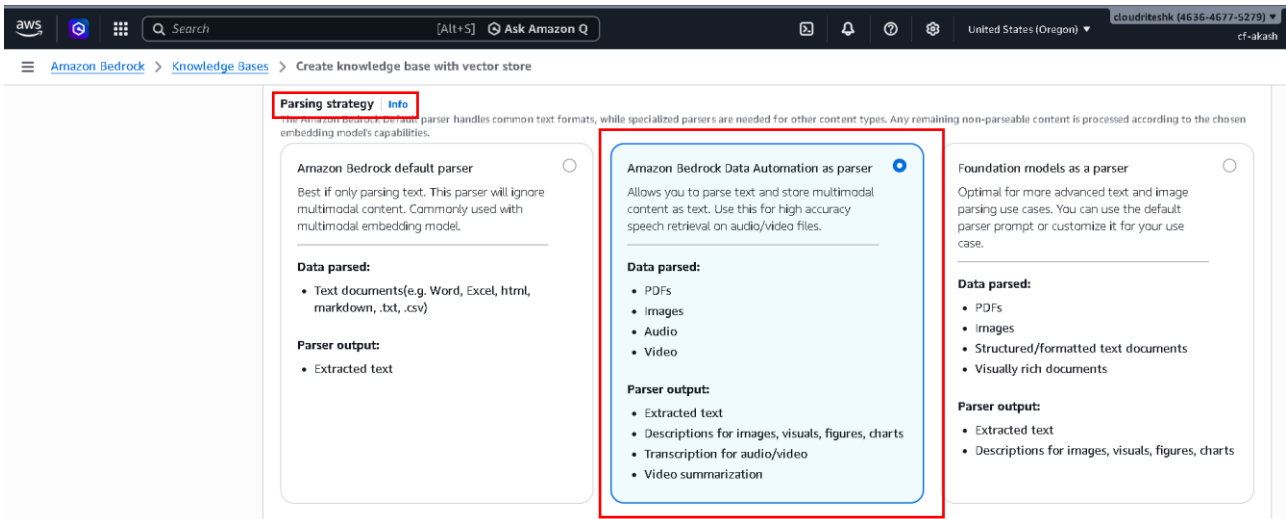
- Click Next.



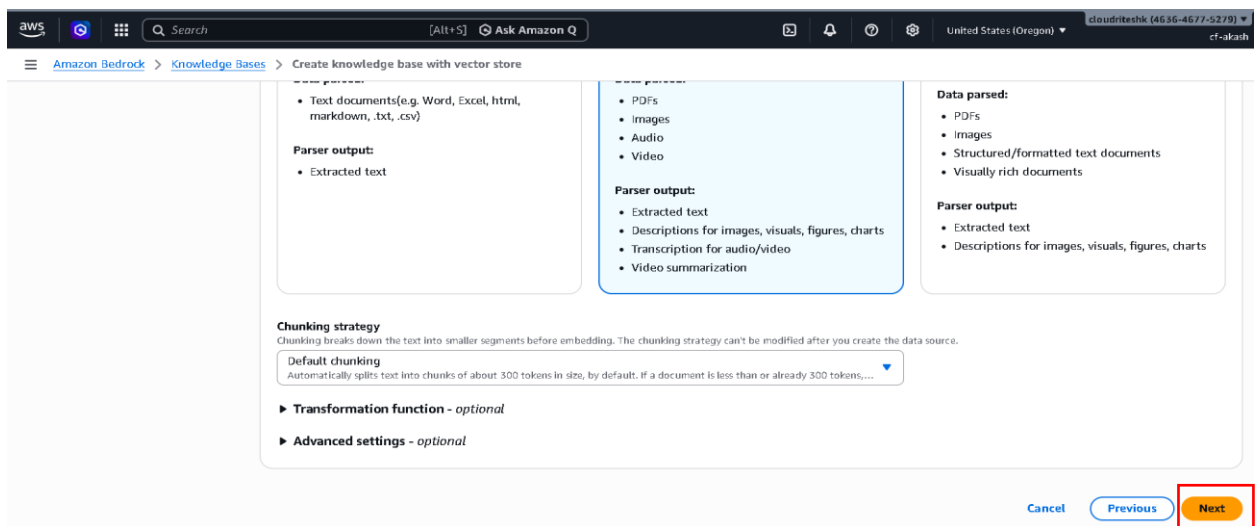
- Write data source name.
- Then Click on Browse S3.



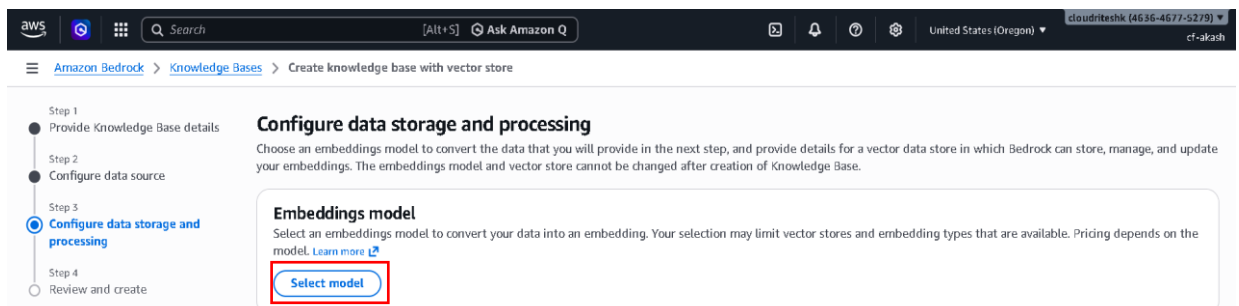
- Choose your created S3 bucket.
- And click on choose.



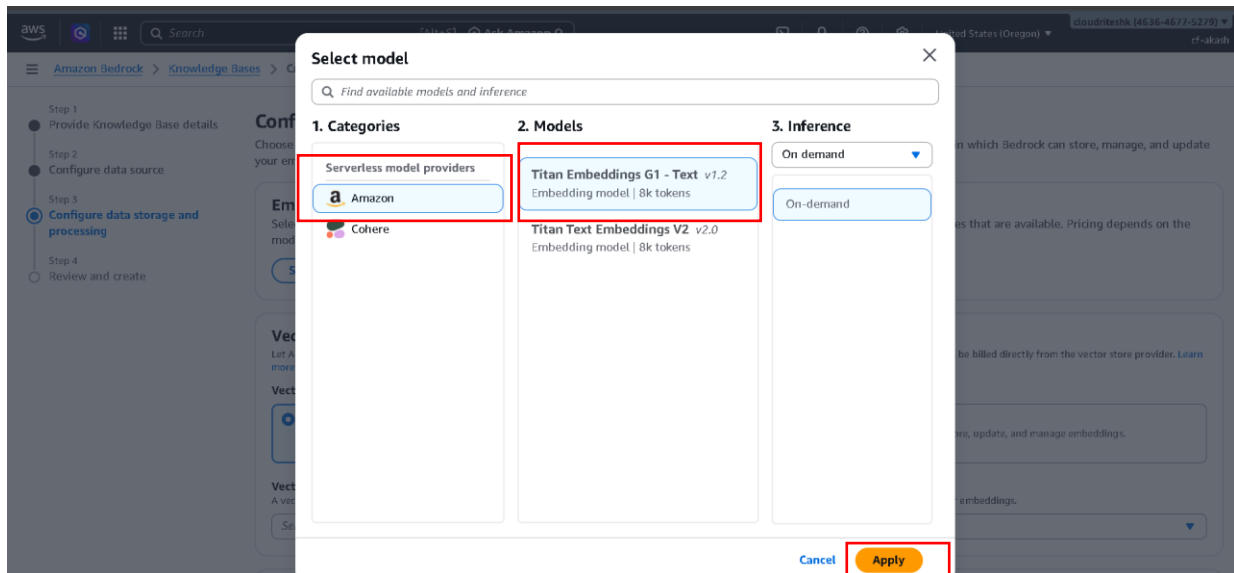
- Choose Parsing strategy
 - Amazon Bedrock Data Automation as parser



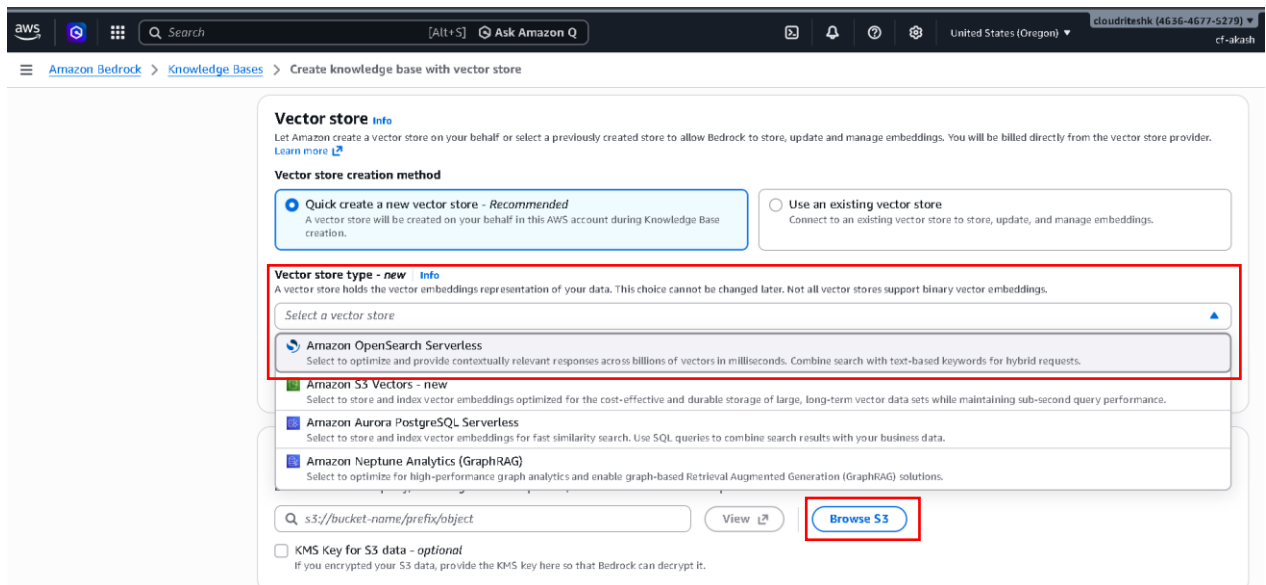
- Click on Next.



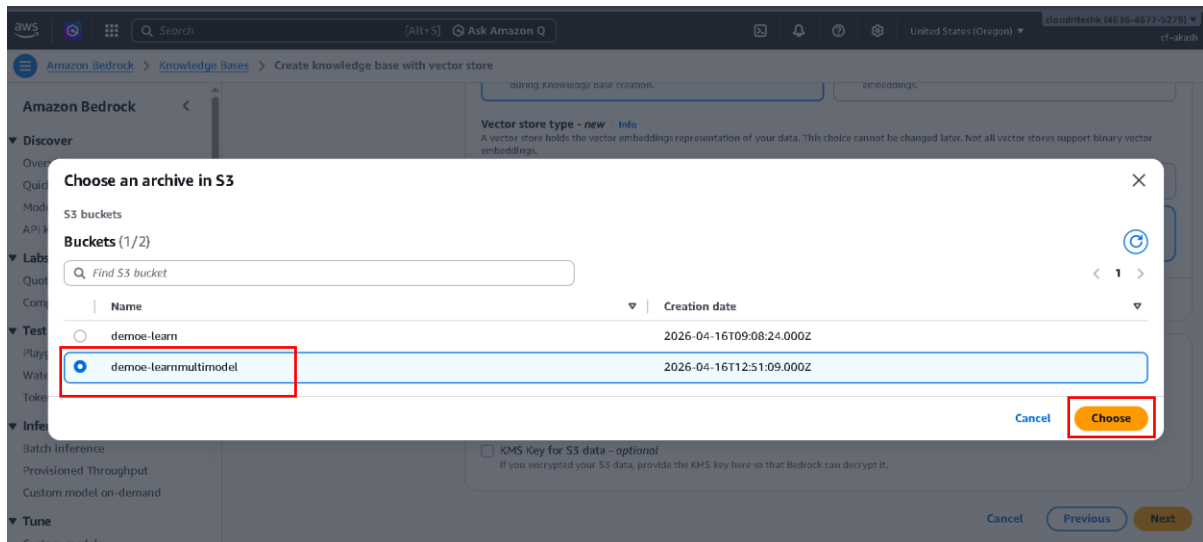
- Click on Select model



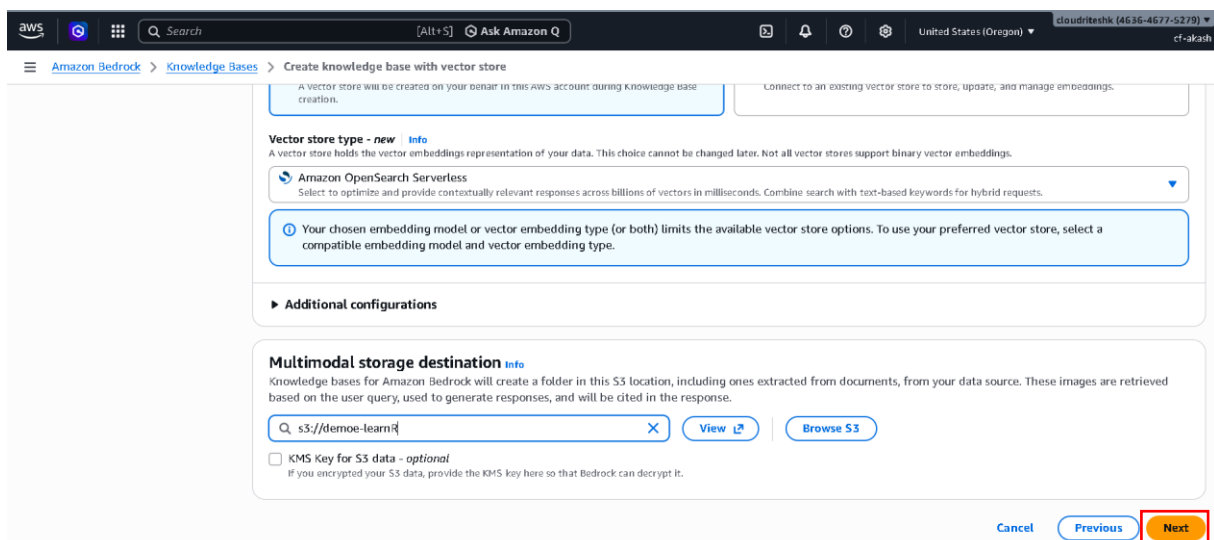
- Choose Amazon as service model provider
- Choose Model v1.2
- And now just click Apply.



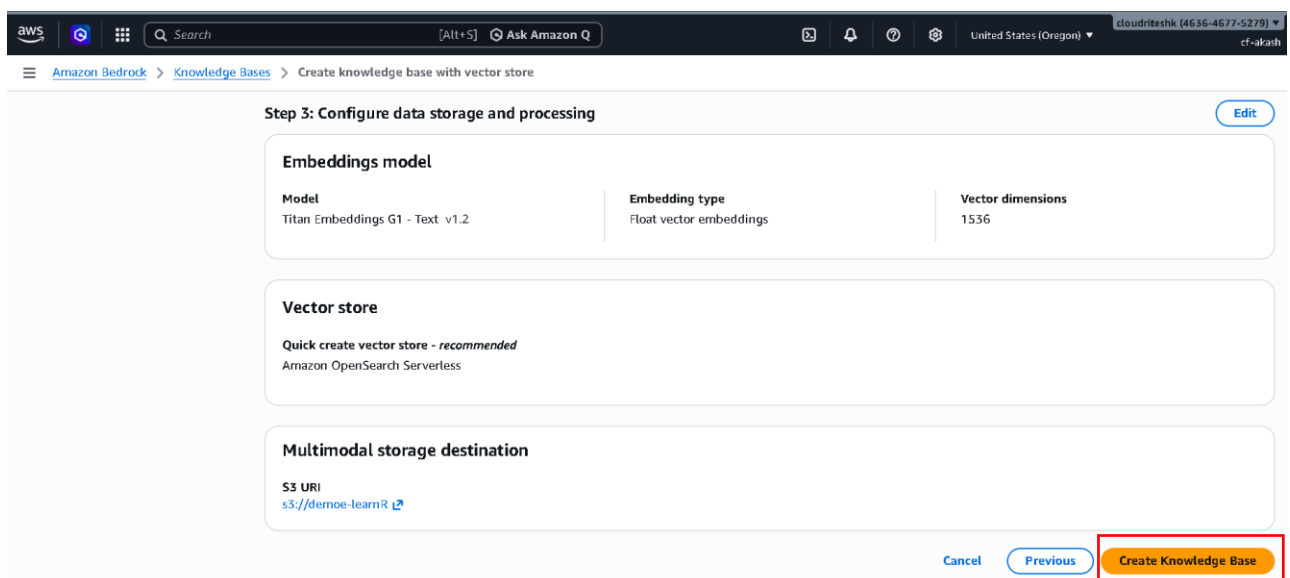
- Choose vector store type
 - Amazon OpenSearch Serverless.
- Click Browse S3.
- Now need one more s3 bucket to store images, video.
- So create new S3 bucket like we create in previous steps



- Select new S3 bucket and click choose



- Now click Next



- Scroll down and click Create Knowledge Base.

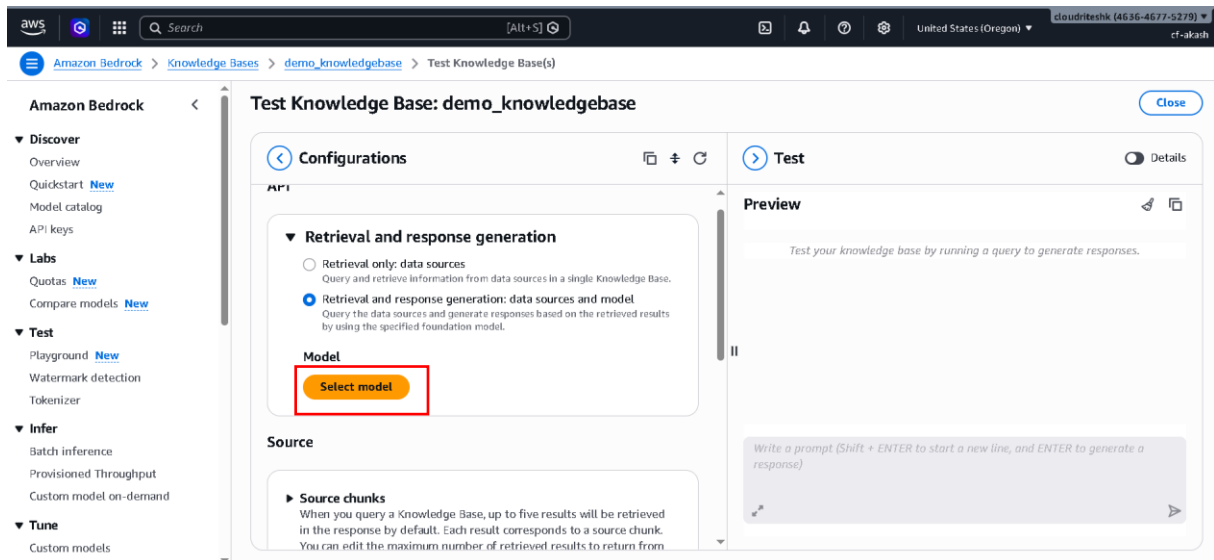
- It take' 5 minute for creation.

The screenshot shows the Amazon Bedrock console interface for a Knowledge Base named 'demo_knowledgebase'. The left sidebar contains navigation options like Discover, Labs, Test, Infer, and Tune. The main content area shows the Knowledge Base overview with details such as Knowledge Base name, ID, description, service role, and status (Available). Below this, the 'Data source (1)' section lists a single data source named 'demo' with a status of 'Available'. The 'Sync' button for this data source is highlighted with a red box.

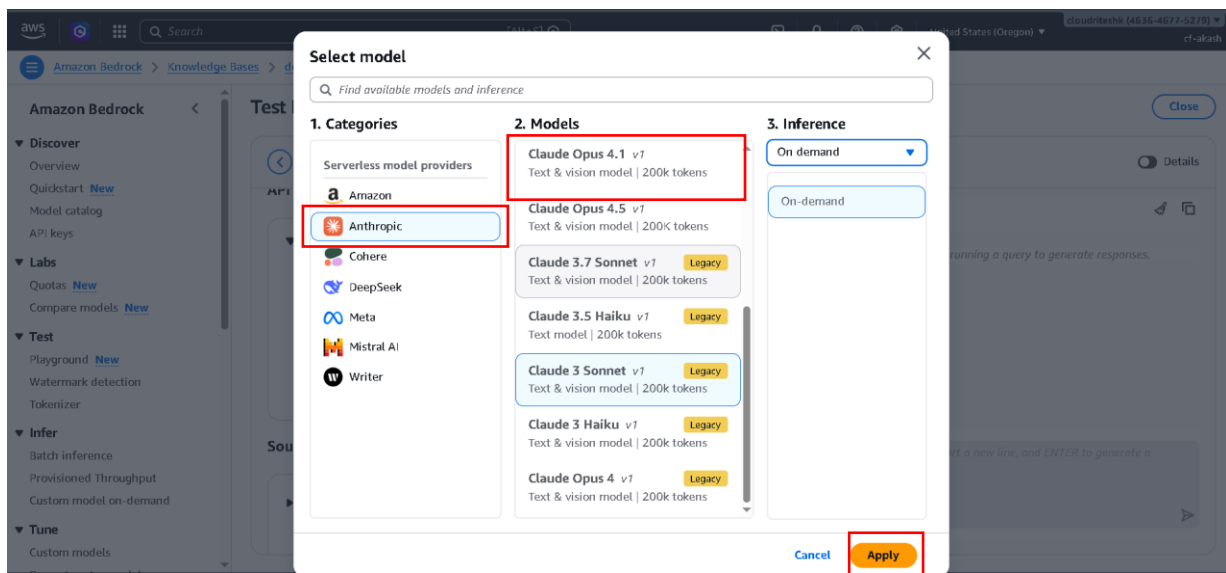
- Now our Knowledge Base is ready.
- Now just select Data source and click sync.
- Syncing time totally depends on your data size.

The screenshot shows the Amazon Bedrock console interface for the 'demo_knowledgebase' page. A green banner at the top of the main content area indicates 'Sync completed for data source - demo'. The 'Data source (1)' section still shows the 'demo' data source. The 'Test Knowledge Base' button is highlighted with a red box.

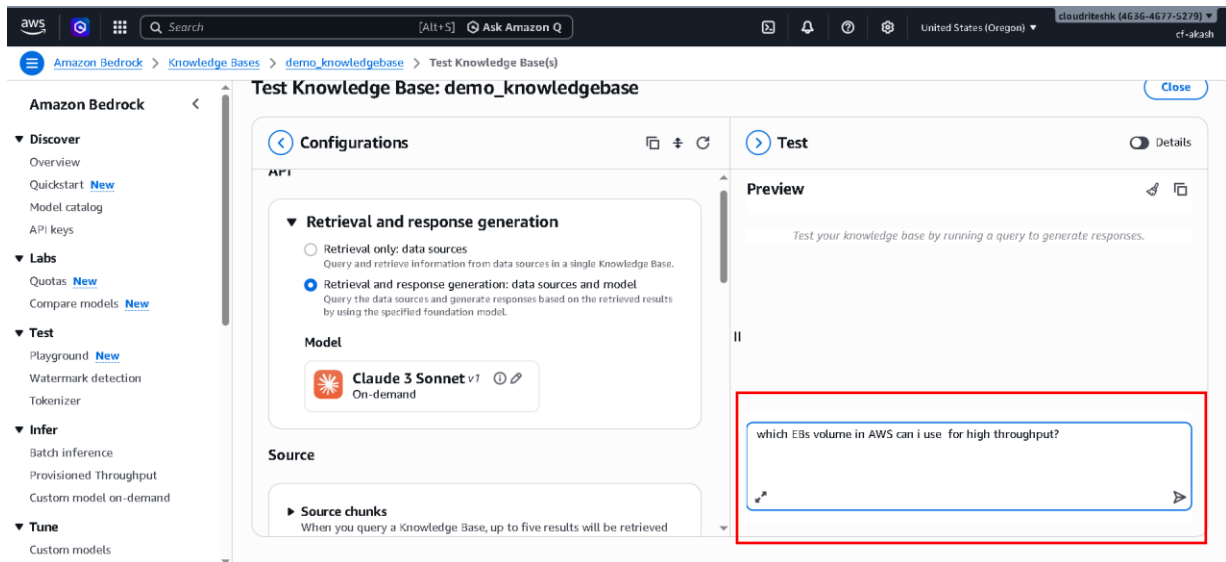
- Sync completed.
- Now just click Test Knowledge Base.



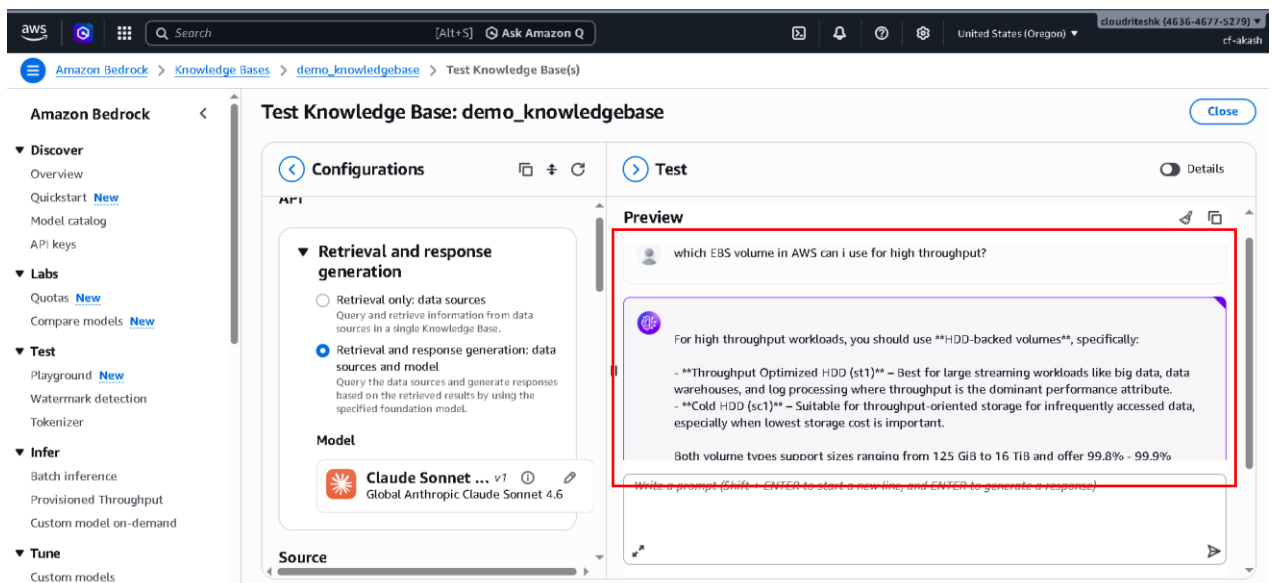
- Now Just click Select model.



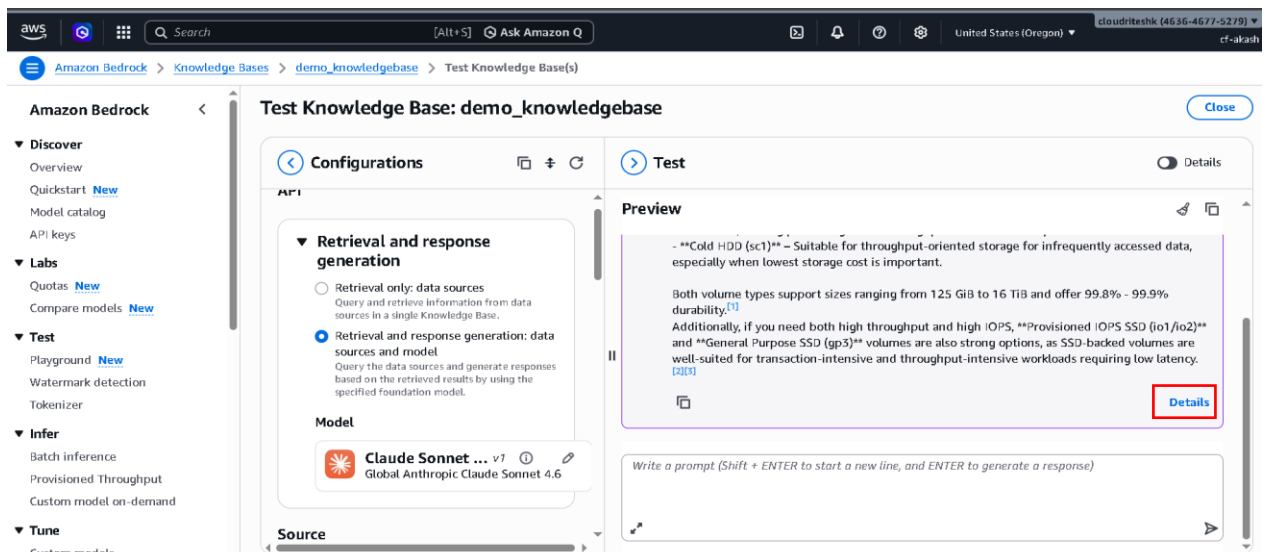
- Select categories and Model.
- And now simply just click Apply.
- A combination of this cloud foundation and data we have in our Knowledge Base is going to be able to respond to our Questions.
- Now ask some Questions.



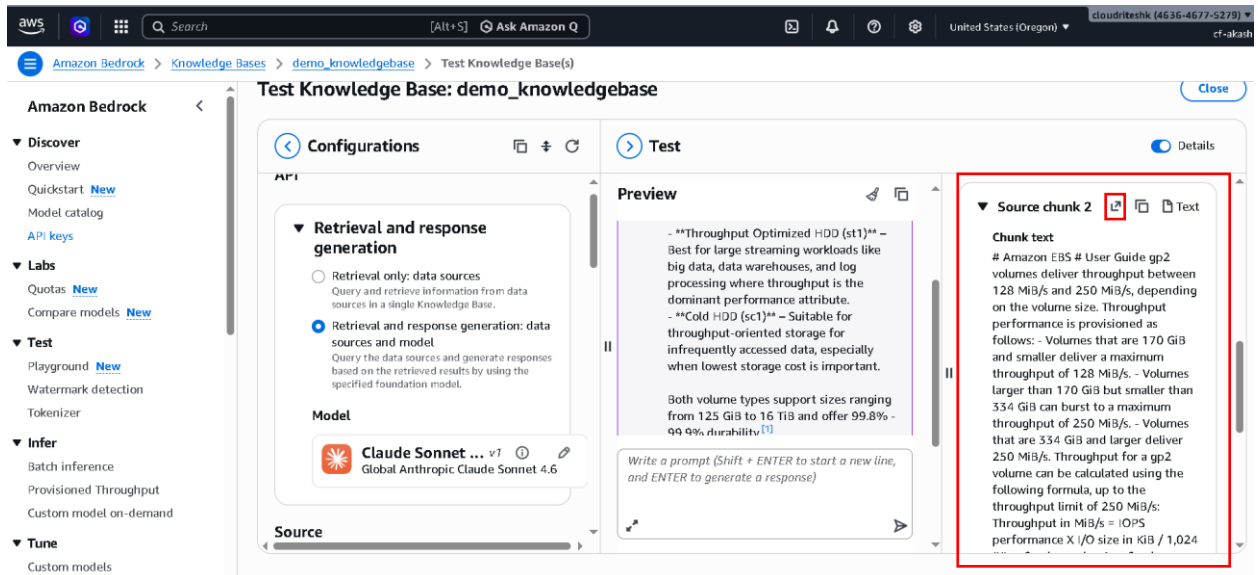
- Now this is the Question I Ask
- Now let's see what they can reply.



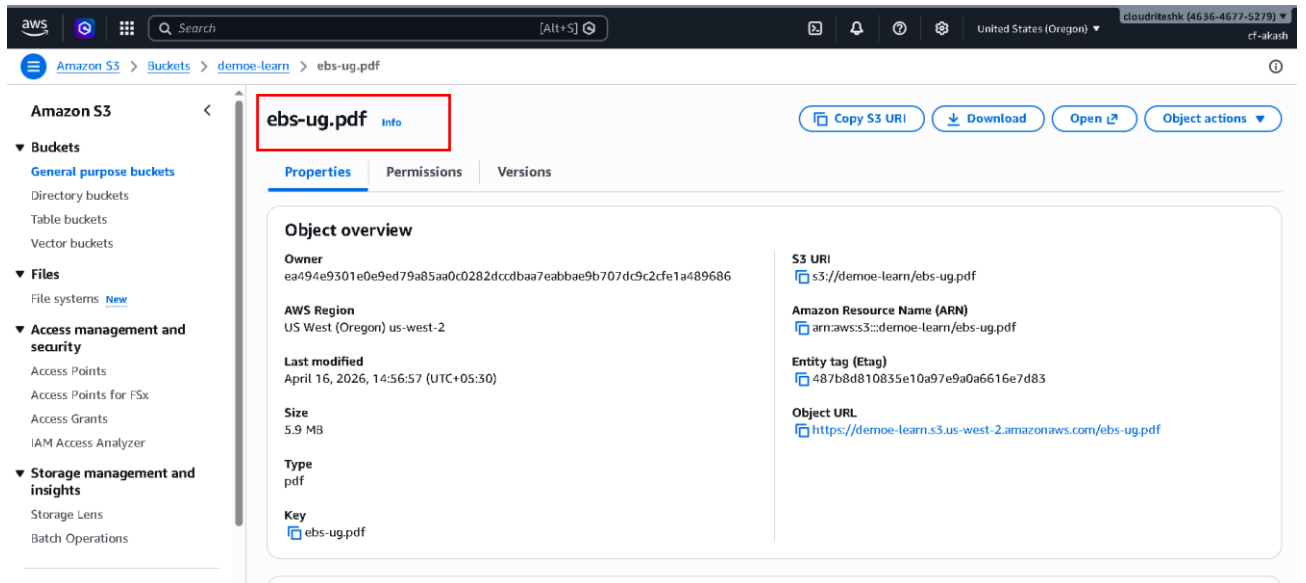
- It's working perfectly.
- Now scroll down.



- Click on Details.



- Here they show this response from this source that we have uploaded in S3 Bucket
- Now click on top first icon in source chunk 2



- So it is show you the source of this response
- So it is ebs-ug.pdf that we had uploaded as our source data.